

CloudStack: An Overview of the Open Source IaaS Platform



Here's an open source IaaS platform to set up an on-demand, elastic cloud computing service. It enables utility computing services by allowing cloud service providers to offer self-service compute instances, storage volumes and networking configurations, and to set up a private cloud for internal use.

Apache CloudStack is an open source, multi-tenanted cloud orchestration platform, which is used to build private, public and hybrid IaaS clouds by pooling computing resources. It manages computing, networking and storage resources. CloudStack is hypervisor agnostic; it uses multiple virtualisation platforms such as KVM, vSphere and XenServer. It supports the Amazon Web Services API, apart from its own APIs.

Features and use cases

CloudStack supports Citrix XenServer, VMware vSphere and KVM on Ubuntu or CentOS. It can manage multiple

geographically distributed data centres. The CloudStack API gives programmatic access to all managed resources and hence it is easier to create command line tools. Multi-node installation support and load balancing makes it highly available. In addition, MySQL replication is also useful for maintaining high availability.

Service providers and organisations use CloudStack to set up

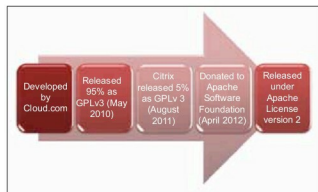


Figure 1: The history of CloudStack



Figure 2: Features